ALAAELDIN ABDELAAL

Full Stack Developer

07776939174 @ alaaeldinabdelaal94@gmail.com https://www.linkedin.com/in/alaaeldinsafwat/

SUMMARY

I'm a software engineer with **5+ years** of hands-on experience specialising in backend and full-stack development. I've designed and delivered end-to-end systems using **Node.js, Go, Python**, and **C#**, taking ownership across the full lifecycle—from system design and implementation to testing, deployment, and production support.

My work includes building secure authentication and authorisation systems, scalable RESTful APIs, and internal automation tools used in production environments. I have strong experience developing web applications with **Angular**, **TypeScript**, and **JavaScript**, and integrating them seamlessly with backend services.

I've worked extensively with **CI/CD** pipelines, unit testing, and secure system design, and I'm comfortable operating in **Linux**-based environments using Docker and Git. Earlier in my career, I worked on **machine learning** and **computer vision** projects with **PyTorch** and **OpenCV**, integrating ML models into backend services for real-world applications.

EXPERIENCE

Software Engineer

GE Vernova

2024 - Present Rugby, United Kingdom

- Developed a device authentication and authorisation system using **Node.js** and **TypeScript**, integrating enterprise identity systems such as **LDAP** and **RADIUS**, and implementing **OAuth2**-based authorisation flows. Built an **Angular** frontend using **HTML** and **CSS**.
- Developed a custom **LDAP** server in **Go**, with supporting **JavaScript** services for user lifecycle management (creation, deletion, authentication).
- Built internal automation tools to update and manage HMI symbols, supporting both GUI and CLI workflows.
- Trained and integrated a **machine learning** model to predict ship draft using environmental data.
- Implemented **CI/CD** pipelines to automate releases, execute unit tests, and perform vulnerability scanning.
- Collaborated with cross-functional teams to deliver secure and reliable systems in production environments.
- Created an **automation tool** to update HMI symbols (aliases, positioning, and transformation handling), accessible via both **GUI and command line**.
- Worked with **React** for front-end development in small to medium features.

Software Engineer

ALTEN - Innovation Lab

2023 - 2024 Derby, United Kingdom

- Developed **RESTful** backend services using **Flask**, **SQLAlchemy**, and **PyTorch** for real-time **object detection** in mobile video streams.
- Built backend systems for 3D CAD model generation from mobile-captured images using **computer vision** and 3D reconstruction techniques.
- Applied **PyTorch** and **OpenCV** to production-oriented **ML** pipelines, integrating models into backend services.
- Worked in an **Agile** environment, contributing to system design, implementation, and testing.

SKILLS

Node JS, Angular, HTML, CSS, RESTful APIs, Sequelize, CI/CD, GO, SQLAlchemy, Linux, Docker, Python, C#, Machine Learning, Deep Learning, GitHub, Agile, Scrum, JIRA, MATLAB, GIT, C++, Embedded systems, PostgreSQL, NumPy, pandas

EDUCATION

MSc in Applied Data Science

University of Central Lancashire

2021

B.Eng. in Mechatronics and Intelligent Machines (1st Class)

University of Central Lancashire

2020

AWARDS

- **ECPC Computer programming contest**
- **3rd place in Egypt in the EAEAT National Competition for Industrial Robotics Applications**
- **IEEEXtreme Competition ambassador**

EXPERIENCE

Software Engineer

University of Central Lancashire

2021 - 2023 Preston, United Kingdom

- Developed software for a 6-DOF industrial robotic system with object detection, implementing inverse kinematics algorithms, control logic, and system integration using **C++, Assembly**, and MATLAB.
- Built a data-driven body stability analysis application, processing EMG and motion data to model the impact of temperature changes on human stability; implemented data pipelines, visualization, and **machine learning** models using **Python**, **Pandas**, **Matplotlib**, **OpenCV**, and **PyTorch**..
- Developed an automated production line system integrating electrical and hydraulic subsystems with a computer vision–based auto-picking mechanism; implemented **PLC** control logic (ladder diagrams), vision processing, and system-level software using **C#** and **C++**.